



AI Freight Agent

The planning brain for global freight.

The planning brain for **global freight**.

Every shipment in the world is a planning decision — what mode, what carrier, what consolidation, what to do when something breaks. Today that brain is fragmented across five mode-specific tools, thirty thousand forwarders, and millions of human-hours.

We are building one engine, rented by any forwarder, that runs the planning loop end-to-end across air, ocean, and trucking.

Tier-1 routing intelligence as infrastructure. We don't replace the TMS. We replace the planner's brain.

PROBLEM

The Tuesday afternoon scene.

A senior air coordinator at a \$200M Tier-2 forwarder sits down at 3pm to build tomorrow's outbound MAWBs. Excel sheet open. WhatsApp threads with HKG / SHA / SIN partner offices. 47 HAWBs in pipeline. Builds three versions over two hours as cargo-readiness shifts, BSA utilization warnings fire, and DG holds drop in late.

No artifact survives the day. Tomorrow they start over.

THE TRUTH

Thirty thousand forwarders globally make decisions like this every day, across every mode. The math is harder than Excel — sequential decisions under cargo-readiness uncertainty, allotment minimums, co-loader relationships, DG / lithium constraints, three coupled cutoffs (CFS, flight, dispatch).

THE IMPACT

Per \$200M forwarder: ~1.5–2 planner FTEs unlocked · cost-to-serve reduction · OTP lift · 80%+ automation of routine decisions. We address the operations-cost pool, not per-seat software spend.

The stack matured this year.

01 STACK

MCP + capable LLMs + open-source MILP solvers + agentic frameworks matured in 2025.

Claude 4.6+, GPT-5, HiGHS, OR-Tools, LangGraph. The same stack would have taken \$20M and 5 years in 2022.

02 PROOF POINTS

Vertical AI is the proven model — different buyer, same trajectory shape.

Harvey: \$50M → \$195M ARR in 12 months at \$11B (legal AI). Augment: \$85M Series A. Raft: 40% of A&A Top-25 on doc AI. We expect 2–3× slower ramp — different unit economics — but services-as-software in vertical AI is established.

03 WINDOW

Tier-1 distraction + TMS dominance = open lane for an intelligence layer.

DSV closed \$15.9B Schenker April 2025 — largest tier-1 distracted for 2–3 years. CargoWise owns 24 of A&A Top-25 — the market wants intelligence on top of the TMS, not replacement.

One engine. Built for the messy parts.

We sit above the TMS — making routing decisions across modes, committing them autonomously when the math is certain AND operator trust is earned.

QUOTE DESK

**Rate request →
cost-to-serve quote.**

Front office.

CONSOLIDATION PLANNER

**Today's shipments →
tomorrow's build plan.**

Network ops. No incumbent at mid-size.

EXCEPTION HANDLER

**Disruption →
customer-aware replan.**

Ops on duty.

THE ENGINE

MILP optimizer + heuristics + ML feasibility predictor + LLM orchestration agent

WHAT THE ENGINE HANDLES THAT EXCEL CAN'T

- Sequential decisions under cargo-readiness uncertainty (re-plan as actuals replace estimates)
- Allotment / BSA minimums + dead-freight penalty math · Co-loader relationship constraints
- DG / lithium / temp-controlled eligibility · Three coupled cutoffs (CFS, flight, dispatch)

Air first. Multimodal next.

WHY AIR CONSOLIDATION FIRST

- Hardest planning math — sequential decisions, density, three coupled cutoffs.
- Highest labor cost per shipment.
- No dominant SaaS incumbent in the consolidation layer.
- Engine extends directly to ocean LCL.

SEED COMMITMENT

- Air consolidation to production at 2 design partners.
- Ocean LCL walking skeleton in trial.
- FCL + Trucking + Intermodal = Series A+ scope.

PHASED BUILD

MVP	Air consolidation planner	SEED
	Production at 2 design partners	
END OF SEED	Ocean LCL consolidation	SEED
	Walking skeleton, first trial	
PHASE 3	Ocean FCL routing	SERIES A
	LaTeX model drafted	
PHASE 4	Trucking routing	SERIES A+
	Closes door-to-door	
PHASE 5	Intermodal stitching	SERIES B+
	Cross-mode tier-1 capability	

Mock UI - consolidation planner - Wed 3pm - LAX

consolidation planner · LAX · 47 HAWBs · Wed 3pm

TOMORROW'S OUTBOUND BUILD

MAWB	DEST	FLIGHT	BUILD	FILL	STATUS
MAWB-001	FRA	LH8404	17 HAWBs · own	82%	ready
MAWB-002	FRA	ECU master	9 HAWBs · co-load	91%	ready
MAWB-003	CDG	AF065	14 HAWBs · own	78%	density var

LH BSA 78% MTD ✓ AF BSA 41% MTD ▲ ECU relationship score 8.4/10

DG / RESTRICTED

HAWB-23 lithium sect II → LH8404 only
HAWB-31 IATA TCR perishable → AF065 only

CUTOFFS

CFS 22:00 · flight 04:30 · dispatch 17:00 today

COST vs BASELINE **-\$1,840 (-7.2%)**

AUDIT TRAIL 12 decisions logged

APPROVE

OVERRIDE — REASON

EXPLAIN MAWB-003

BSA utilization, DG eligibility, audit trail — every override logged with reason. The engine handles the constraints Excel cannot.

The moat is the math layer.

01 THE MOAT

Explainable math · auditable decisions · earned autonomy.

LLM-only competitors cannot certify their answers — they need humans to approve every booking. Our engine carries a certificate of optimality, an explanation tree per decision, and an immutable audit trail. The agent commits autonomously only when math is certain AND operator trust is earned.

02 BUY-VS-BUILD

We are competing with the customer's internal data team.

Tier-2 forwarders with internal DS can build 60–70% of this engine in 12–18 months for ~\$1.5–2M. We get them to production in 16 weeks for a \$25K paid pilot. Time-to-value, not feature parity, is the arbitrage.

03 THE EDGE

Built this architecture in tier-1 platforms — twice.

\$110M+ annual savings · 82% automation · 72% cost-to-serve reduction. We are packaging what only tier-1s could afford to build.

TAM is the revenue we orchestrate.

\$216B

TAM · GLOBAL FORWARDING REVENUE

Armstrong & Associates 2024.

\$40–120M

5-YR SOM · ARR

Base case \$40–80M; bull \$100–150M with outcome pricing.

BOTTOM-UP SAM BUILD — DEFENSIBLE MATH

Air consolidation labor	\$400K / \$200M forwarder	1.5–2 senior planners @ \$200K fully-loaded
Multiply across modes	~\$2M / \$200M forwarder	+ ocean LCL/FCL + trucking + quote desk + exception
Penetration	30% of tier 1+2 globally	~750 forwarders × \$2M = \$1.5B addressable now
Wallet expansion (5yr)	\$4–6B SAM	Tier 3 enters via channel; AI layer matures
SOM capture	1–3% of SAM	Tier-2 wedge + narrow tier-1 + early channel

Land Tier 2. Expand both ways.

TIER 2 \$100M – \$5B revenue ~3,000 globally STAGE · SEED LANDING ZONE. Full intelligence layer. Direct sales. Pain + budget + no in-house OR. Why land here: tier-1 distraction window + tier-2 squeeze. ACV \$30K – \$100K ACV	TIER 1 >\$5B revenue ~25 globally STAGE · SERIES A+ Narrow-capability overlay alongside CargoWise. Sell the planning brain, not the TMS. Raft precedent: 40% of A&A Top-25 on document AI. ACV \$250K – \$1M+ ACV	TIER 3 \$5M – \$100M revenue Tens of thousands STAGE · SERIES B+ Channel via TMS marketplaces + customs-broker network. Self-serve. CargoWise's 17K-customer tail proves channel economics. ACV \$1K – \$5K / month
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Seed = land Tier 2. Tier 1 and Tier 3 are post-traction expansions over 5–7 years.

The real competition is buy-vs-build.

Player	What they do	Reality check
CargoWise / WiseTech	TMS system of record · 24 of Top-25	We sit on top. Primary risk: native multimodal planning extension.
Augment, Raft	Inbox / doc AI · \$85M Series A; 40% Top-25	Distribution we don't have. Likely partnership or acquisition target.
cargo.one, WebCargo, Freightos	Air rate API + booking	Different layer. Rate plumbing under our quote desk.
Flexport, Forto	Digital forwarder operators	They became forwarders. Forto pivot vindicates SaaS shift.
p44, FourKites, Beacon, Logixboard, Shipamax	Visibility / ETA / doc OCR	Different layer — feed our engine as data sources.
Tier-1 in-house (Tango, eTouch, MyDHLi)	Proprietary planning brains	Won't license. Proof the architecture works.
Customer's in-house DS team	Build 60–70% in 12–18mo for \$1.5–2M	THE REAL COMPETITION. We win on time-to-value, not feature parity.

THREE COMPETITIVE SCENARIOS TO WATCH

- 01 WiseTech extends CargoWise into multimodal planning. Medium probability, 18–24mo. PRIMARY RISK
- 02 Augment / Raft moves up-stack into routing decisions. Medium probability, 12–18mo with their capital.
- 03 A tier-2 builds internally and shares (open-source). Low probability; threat shapes our pricing.

Sequenced distribution. Buyer-defined KPI.

DISTRIBUTION PATHS — HONEST ABOUT TIMELINES

Path	Time to first customer	Why
GoFreight accelerator	60–90 days	Open REST API, no partner gate, mid-size customer base
CargoWise via eAdaptor	9–14 months / customer	Partner application Q3 2026 + per-customer integration
Magaya / Riege	12+ months	After design-partner traction

DESIGN PARTNER CONTRACT — BUYER-DEFINED KPI

- \$25K paid pilot · 16 weeks · scope: air consolidation planner on one gateway
- Primary KPI: planner hours saved per build cycle vs pre-pilot baseline (the buyer's metric)
- Secondary: co-load mix shift · cost delta vs baseline · audit completeness
- Go/no-go gate at week 12 · refund clause if KPI not met · production conversion at 90 days

TRUST RAMP — JOINTLY DEFINED, NOT VENDOR-SET

Co-pilot · ≥4 weeks

Supervised · ≥8 weeks

Autonomous per-lane · earned

Richard Li-Yang Chen

Founder & CEO · Built this architecture inside Tier-1 freight platforms — twice.

\$110M+

ANNUAL SAVINGS

82%

AUTOMATION OF PRIOR MANUAL PLANNING

72%

COST-TO-SERVE REDUCTION

Top-5 global digital freight forwarder · 2021–2023

Director, PDS / Global Head of Applied Science. Multimodal planning across ocean FCL/LCL, air, drayage, trucking, consolidation. Built ML transit-time estimator. Scaled AS org 3 → 25+.

Top-10 global e-commerce platform · 2024–present

Director, Supply Chain Tech. Leads 30+ engineers / DS / PMs. S&OP/IBP + demand forecasting + multi-vendor OMS 0-to-1 in 6 months.

Fortune 50 e-commerce + logistics platform · 2018–2019

Research Science Manager, Middle-Mile. Linehaul routing — 100,000+ concurrent loads daily. Cross-border China → US/EU.

U.S. Department of Transportation · 2024

Special Advisor, OMFIP. National Multimodal Freight Modeling System. Supported White House Council on Supply Chain Resiliency.

Sandia National Laboratories · 2013–2017

Principal Member of Technical Staff. Stochastic optimization research; top-tier OR journals.

EDUCATION **Ph.D. Operations Research, University of Michigan** · M.S. + B.S. IEOR, U.C. Berkeley

IP POSTURE Clean. No non-compete on this product domain. Architecture is general OR + agentic AI; no copied artifacts.

Founder thesis. Built this for tier-1 platforms twice. No one packaged it as a product. The window is now.

Where we are. What we need.

WHERE WE ARE · MAY 2026

- 01 Phase 0 complete — PRD v0.3; 4 LaTeX models drafted (Air, Ocean FCL/LCL, Trucking).
- 02 Air model under PDF review; walking-skeleton MILP code passing 5 isolation tests.
- 03 Primary research mapped 18 operational roles — consolidation planner is the only role without product coverage.
- 04 [N] forwarder intro calls; [M] in active conversation. Named pipeline in DD packet.

RAISE

\$2.5M

seed · 14-month runway



OUTCOME-BASED MILESTONES — ENTERPRISE-READY

M6

2 design partners signed. Planner live in shadow mode. SOC 2 Type I scoped.

M9

Planner hours saved ≥20% on first lane. Reference customer agreement secured.

M14

3 production deployments. First annual. Ocean LCL in trial. Series A-ready.